



Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 5

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Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	11/7/2026
Title of Assignment:	Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.
Problem Statement	Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.
Learning objectives	To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.
Theory	ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory

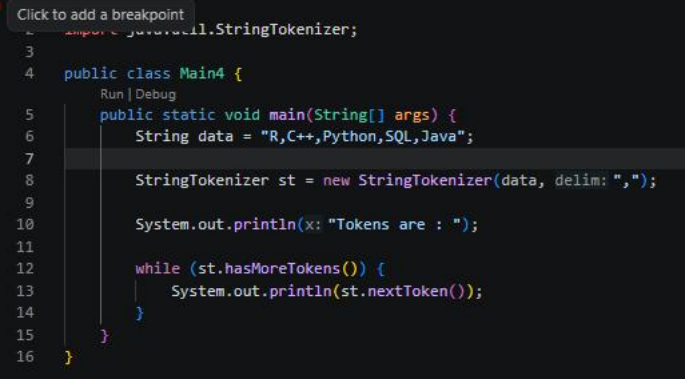
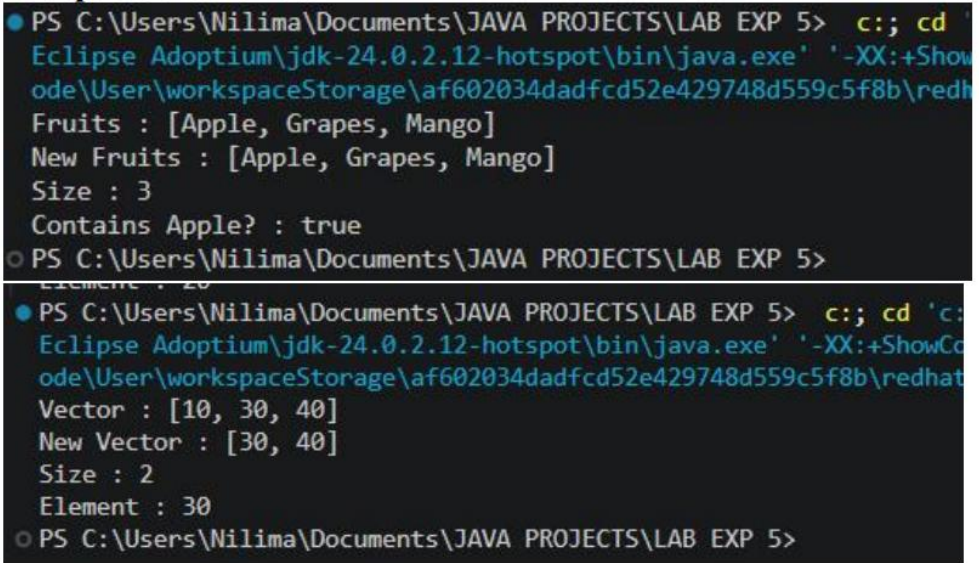
allocation and efficient string processing

Program Code

```
1
2 import java.util.ArrayList;
3
4 public class Main1 {
5
6     public static void main (String[] args) {
7         ArrayList<String> arr = new ArrayList<>();
8
9         arr.add("Apple");
10        arr.add("Grapes");
11        arr.add("Mango");
12
13        System.out.println("Fruits : " + arr);
14        arr.remove("Banana");
15        System.out.println("New Fruits : " + arr);
16        System.out.println("Size : " + arr.size());
17        System.out.println("Contains Apple? : " + arr.contains("Apple"));
18    }
19 }
```

```
1
2 import java.util.Vector;
3
4 public class Main2 {
5     public static void main (String[] args) {
6         Vector<Integer> arr = new Vector<>();
7
8         arr.add(10);
9         arr.add(30);
10        arr.add(40);
11
12        System.out.println("Vector : " + arr);
13        arr.remove(0);
14        System.out.println("New Vector : " + arr);
15        System.out.println("Size : " + arr.size());
16        System.out.println("Element : " + arr.get(0));
17    }
18 }
```

```
1
2 public class Main3 {
3     public static void main(String[] args) {
4         StringBuffer sb = new StringBuffer(str: "Programming");
5
6         System.out.println("Original String : " + sb);
7
8         sb.append(str: " in Java");
9         System.out.println("After Append : " + sb);
10
11        sb.insert(offset: 12, str: " Language");
12        System.out.println("After Insert : " + sb);
13
14        sb.replace(start: 0, end: 11, str: "Coding");
15        System.out.println("After Replace : " + sb);
16
17        sb.delete(start: 6, end: 15);
18        System.out.println("After Delete : " + sb);
19
20        sb.reverse();
21        System.out.println("After Reverse : " + sb);
22    }
23 }
```

	 <pre> 1 import java.util.StringTokenizer; 2 3 4 public class Main4 { 5 public static void main(String[] args) { 6 String data = "R,C++,Python,SQL,Java"; 7 StringTokenizer st = new StringTokenizer(data, delim: ","); 8 9 System.out.println(x: "Tokens are : "); 10 11 while (st.hasMoreTokens()) { 12 System.out.println(st.nextToken()); 13 } 14 } 15 } 16 </pre>
Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'C:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redh Fruits : [Apple, Grapes, Mango] New Fruits : [Apple, Grapes, Mango] Size : 3 Contains Apple? : true PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> Element : 20 PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'C:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redh Vector : [10, 30, 40] New Vector : [30, 40] Size : 2 Element : 30 PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> </pre>

```

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> &
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Nilima
edhat.java\jdt_ws\LAB EXP 5_8277c32d\bin' 'Main3'
● Original String : Programming
  After Append   : Programming in Java
  After Insert   : Programming Languagein Java
  After Replace  : Coding Languagein Java
  After Delete   : Codingein Java
  After Reverse  : avaJ niegnidoC
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>

```

```

JavaScript
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c++;
Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+
ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\
Tokens are :
R
C++
Python
SQL
Java
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5>

```

Exercise Qns

Q1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

```

1
2 import java.util.ArrayList;
3
4 public class Ex1 {
5     Run | Debug
6     public static void main (String[] args) {
7         ArrayList<String> arr = new ArrayList<>();
8
9         arr.add(e: "Nap");
10        arr.add(e: "Build a project");
11        arr.add(e: "Study");
12        arr.add(e: "Practice problems");
13        arr.add(e: "Coding");
14        arr.add(e: "Review notes");
15        arr.add(e: "Play a sport");
16        String temp = arr.get(index: 1);
17        arr.set(index: 1, arr.get(index: 3));
18        arr.set(index: 3, temp);
19
20        StringBuffer str = new StringBuffer();
21
22        for (int i = 0; i < arr.size(); i++) {
23            str.append((i + 1) + ". " + arr.get(i) + "\n");
24        }
25
26        System.out.println(x: "----- TO DO List -----");
27        System.out.println(str);
28    }
29 }

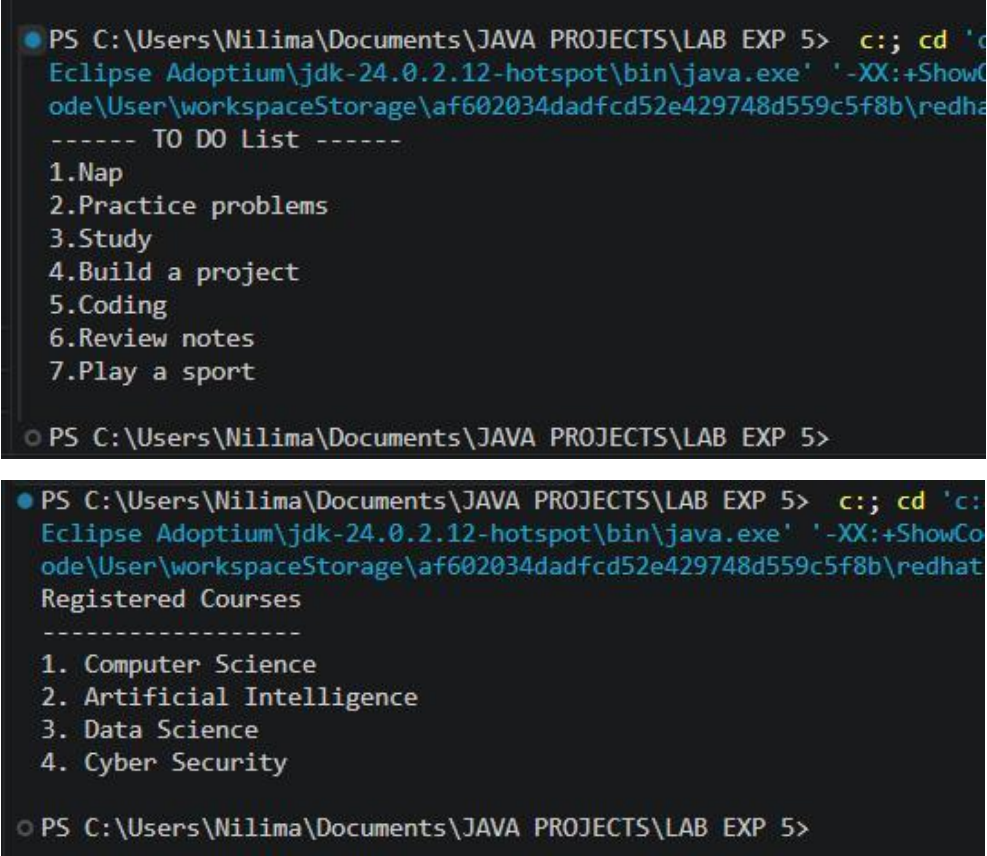
```

Q2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```

1
2 import java.util.ArrayList;
3
4 public class Ex2 {
5     Run | Debug
6     public static void main(String[] args) {
7         ArrayList<String> courses = new ArrayList<>();
8
9         courses.add(e: "Computer Science");
10        courses.add(e: "Artificial Intelligence");
11        courses.add(e: "Data Science");
12        courses.add(e: "Cyber Security");
13
14        courses.remove(o: "Java");
15
16        StringBuffer sb = new StringBuffer();
17
18        sb.append(str: "Registered Courses\n");
19        sb.append(str: "-----\n");
20
21        for (int i = 0; i < courses.size(); i++) {
22            sb.append((i + 1) + ". " + courses.get(i) + "\n");
23        }
24
25        System.out.println(sb);
26    }
27 }

```

Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'c:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redhat' ----- TO DO List ----- 1.Nap 2.Practice problems 3.Study 4.Build a project 5.Coding 6.Review notes 7.Play a sport PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> c:; cd 'c:\Eclipse Adoptium\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCode\ode\User\workspaceStorage\af602034dadfcd52e429748d559c5f8b\redhat' Registered Courses ----- 1. Computer Science 2. Artificial Intelligence 3. Data Science 4. Cyber Security PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 5> </pre>
Conclusion	This assignment helped in understanding the use of ArrayList, Vector, and StringBuffer in Java for dynamic data storage, management, and string manipulation. It demonstrated how ArrayList and Vector can efficiently store and manage collections of data, while StringBuffer enables efficient modification of strings without creating new objects. The implementation improved my understanding of the Java Collections Framework, dynamic memory management, and mutable string operations. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, database management systems, and other real-world Java applications that require efficient data handling and string processing.